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BACHELOR THESIS IN COMPUTER SCIENCE

Extracting decision trees for clinical guideline-based chatbots

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Abstract

This report investigates how NICE clinical guidelines can be transformed into formal, machine-readable structures while preserving their original clinical reasoning. Using a set of ten NICE visual guideline summaries, we develop a pipeline that first extracts rule-based representations and then reconstructs them into decision trees. Four language and vision-language models are compared for this task, with MiniVision producing the most faithful and coherent structured outputs overall. The extracted trees are then extended for downstream use by generating synthetic patient cases, supporting situations with missing patient information, and annotating outcomes with urgency levels. Results show that clearly structured flowcharts can be transformed with high fidelity, while visually dense or irregular formats remain more challenging. Overall, this work demonstrates that NICE guidelines can be converted into interpretable computational decision structures that provide a promising basis for future clinical decision support and guideline-based conversational systems.

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1 Introduction

The development of AI-powered health chatbots aims to build conversational agents able to assist clinicians. As such, it holds the exciting promise to improve healthcare efficiency, availability and unbiasedness (Ive et al., 2025). However, generation of clinical recommendations by LLMs presents safety risks, most notably hallucinations and ungrounded outputs (Krešević et al., 2024). As these challenges are a serious concern in clinical environments, a pragmatic direction is to use LLMs that prioritize evidence retrieval of validated guideline sources.

Clinical guidelines translate empirical clinical trial and expert consensus into actionable recommendations for diagnosis, management, and escalation. They are a cornerstone of evidence-based healthcare decision support. For instance, in the UK, national guidelines from the National Institute for Health and Care Excellence (NICE) provide widely used standards for clinical practice and health technology adoption. But despite their centrality to safe practice, guideline use in real clinical care poses a myriad of issues. Indeed, high time pressure and complex documentation result in incomplete adherence and variability in care (Ive et al., 2025). There is therefore a strong motivation to develop clinical decision support systems (CDSSs) that make guideline-aligned actions easier to access for clinicians.

In this paper, we investigate how NICE clinical guidelines can be systematically transformed into formal structures suitable for computational use while preserving the original reasoning. By formalizing guideline knowledge in this way, the resulting representations can support downstream clinical decision support systems and conversational interfaces. The broader objective is to enable reliable and interpretable guideline-based decision support that remains faithful to validated clinical recommendations.

2 Background and related work

2.1 Health chatbots

Health chatbots are AI systems designed to simulate conversations with patients and clinicians. They use natural language processing (NLP) techniques and machine learning to interpret user inputs and generate contextually appropriate responses. Their interfaces are typically text or voice based, with recent models also being embedded in apps, web portals, and telehealth platforms. They support a broad range of applications in healthcare, amongst which we will be focusing on decision-making support for various medical conditions. The appeal of chatbots lies in their scalability and availability: they can offer constant assistance, reduce the burden on clinical staff, and provide timely responses. In settings with limited resources, they can act as support systems; ultimately helping guide patients toward appropriate care.

On a side note, it has been observed that “generic” chatbots underperform in healthcare against domain-adapted language models. Indeed biomedical contexts have their own vocabulary, abbreviation patterns, and structure. And so models such as BioBERT (pretrained on biomedical corpora), and ClinicalBERT models (adapted to clinical text, including discharge summaries and clinical notes) have shown that specific pretraining improves performance on downstream clinical NLP tasks. In parallel, practical deployment pushes toward smaller and faster models. DistilBERT for instance demonstrates that knowledge distillation can substantially reduce compute and latency while retaining most of the parent model’s language understanding, which is attractive in low-resource or real-time settings.

It is really important to note that clinical deployment raises higher standards for correctness and traceability than general-purpose Q&A. Moreover responses can be sensitive to prompt phrasing, user demographics, or linguistic bias (Dušek, Novikova and Rieser, 2020). In high stakes fields like healthcare, this can plausibly lead to patient harm. The hepatology guideline interpretation study (Krešević et al.) explicitly frames inaccurate responses (“hallucinations”) as a major barrier to using LLMs for clinical guidance. It motivates techniques that bind model output to guideline content. Another argument is the importance of faithfulness and auditability. Patient responsibility in clinical contexts requires the ability to retrace answers back to the exact lines of evidence. We will thus focus on using LLMs primarily for selection and extraction of relevant guideline text, and similarly to previous work done by the UCL team (Institute of Health Informatics) confine outputs to verbatim guideline material to reduce hallucination risk.

2.2 NICE clinical guidelines

NICE is an executive non-departmental public body of the Department of Health and Social Care of the UK. It publishes guidelines across multiple domains, including clinical practice guidelines and health technology assessment for medicines and interventions, whose appraisals are based on evidence-based clinical and cost-effectiveness. This means it provides guidance on decisions that will maximize both desired medical outcomes and economic arguments, based on cost per quality-adjusted life year (QALY) and incremental cost-effectiveness ratios (ICERs).

We will be basing our work on the following official clinical guidelines:

- Hypertension
- Pregnant women at risk of pre-eclampsia
- Preventing relapse in depression
- Antimicrobial prescribing in acute otitis media

- Antimicrobial prescribing in acute sore throat
- Management of chronic open angle glaucoma
- Management of ocular hypertension
- Antimicrobial prescribing in recurrent UTI
- Imaging following head trauma
- Antimicrobial prescribing human and animal bites

NICE guidelines are made by groups consisting of medical professionals, representatives of patient and carer groups and technical experts, and they go through thorough review. Nevertheless, it is important to note that these recommendations are meant to support and not replace clinical decision-making. Moreover, it is common for UK hospitals to additionally refer to local guidelines that incorporate region specific information such as antimicrobial resistance patterns, referral pathways, patient medical history, and contact details for specialist teams. Finally, it has been noticed that national guideline retrieval in practice is bottlenecked by search cost, navigation cost, and interpretation cost (mapping a patient-specific query to the appropriate recommendations). These issues are particularly prevalent in contexts of high time pressure and patient case complexity (Ive et al., 2025).

We will therefore focus on building a tool for information access and guideline navigation, making use of a high-reliability source which is underexploited due to clinical constraints. The goal is to have conversational retrieval systems that reduce time-to-answer while keeping responses bounded to guideline content, and ultimately have local guidelines complement NICE-level resources by adding region-specific and operational information.

2.3 Processing guidelines

The project relies on how we adapt guidelines into a representation that is retrieval-friendly and reasoning-friendly. The following two papers are relevant here, because they show why formatting choices dominate, and how flowcharts can be converted into decision structures that LLMs can actually execute.

Krešević et al. propose a guideline-aware framework for chronic HCV management using GPT-4 Turbo with retrieval augmented generation (RAG) and prompt engineering. They essentially run an ablation study on “how much does the model improve as we make the guideline easier to consume?”. Their key finding is that data quality beats data quantity: baseline accuracy is poor (43%), and improves stepwise when guidelines are (i) provided in-context, (ii) cleaned, (iii) converted from non-text elements into text, and (iv) forced into a consistent structure with prompt templates that match that structure; overall reaching 99% accuracy. Krešević et al. explicitly point out that variation in guideline format affects interpretation and retrieval quality, which is exactly the problem NICE visual summaries raise for an LLM pipeline. Thus we should expect gains from building a pipeline that normalises structure and machine-readable conversions.

MedDM introduces an LLM-executable Clinical Guidance Tree (CGT), where nodes are expressed in natural language and the structure is represented as a multi-branch tree that an LLM can traverse step-by-step. Indeed, guidelines often encode decision logic as flowcharts, but LLMs do not automatically “execute” flowcharts reliably. They thus build a dataset by extracting flowcharts from medical literature and converting them into CGTs: layout analysis to extract figures, shape detection with Faster R-CNN, connector recognition using contour detection and DBSCAN clustering, OCR for node text (CnOCR), and then reconstruction and manual calibration to ensure the node content and edges are correct. This also allows for a multi-turn process. In fact, decision tree structures have practical advantages for clinical CDSS: (i) instead of free-form generation, the model is

constrained to a finite set of guideline-defined branches and outcomes, (ii) trees produce a readable and traceable “path explanation”, which is similar to how clinicians make decisions, (iii) trees naturally support patient conditional contexts and clarification dynamics.

These works serve as concrete starting points and suggest that our technical challenge is centered around representation: extracting clinical decision trees from NICE guidelines and integrating them into our guideline-based chatbot pipeline.

3 Bijective transformation

This section outlines our main computational work. The central aim is to convert NICE guideline content into formal decision structures while preserving the original clinical reasoning. To do so, we represent guidelines as explicit rules and then reconstruct these rules into decision trees. This transformation is intended to be bijective, in the sense that the initial guidelines and the resulting trees should remain mutually consistent and convey the same decision logic. Establishing this correspondence is important both for faithful computational use of guidelines and for assessing whether the extracted structures accurately reflect the source material.

3.1 Guideline to rules

NICE guidelines are typically provided as visual summaries or algorithmic flowcharts combining short text blocks, arrows, thresholds, and branching conditions. A single page may therefore contain several decision points, treatment options, and exceptions arranged spatially rather than as continuous prose. The example below is the first algorithm of a guideline for imaging following head trauma. We will be presenting the transformations based on this one, the rest of the extracted decision trees can be found in the appendix (A.2).

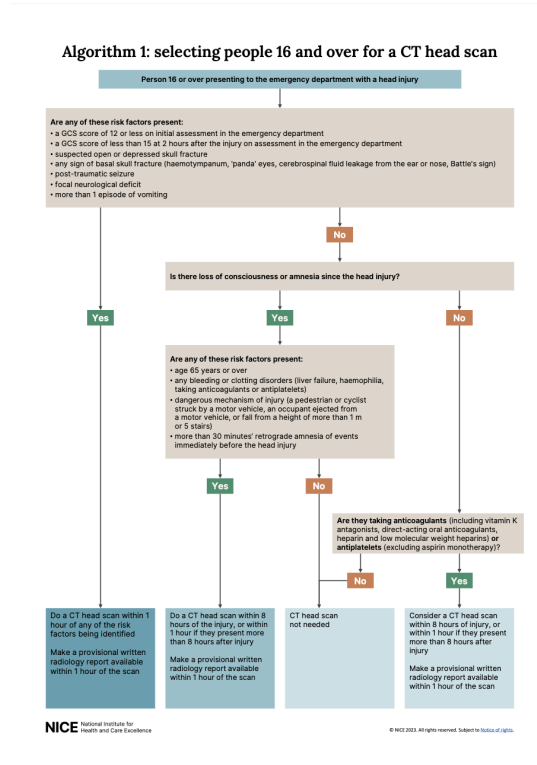


Figure 1: Original NICE visual summary for selecting people aged 16 and over for a CT head scan.

In this step, we transform such guideline material into an explicit rule format that could be processed computationally. The extracted rules take the form of structured IF–THEN statements.

We first explore non-vision models, hoping to base our transformation on the textual content of the original guidelines. For this, we first extract the textual content of the PDF guidelines and use it as the input to a structured information extraction pipeline. The extracted text is passed to an LLM with a predefined JSON schema specifying the fields to be returned. This includes guideline metadata, decision rules, nodes, and edges. Crucially, the model is used for extraction rather than open-ended generation, so that it reorganizes guideline content already present in the document instead of producing new clinical recommendations. The resulting JSON

object is then parsed for reconstruction into explicit rules. In this step, we compare two models: MiniGPT4 and Llama. They are chosen as baselines because they are able to process long textual inputs while remaining relatively lightweight and practical for experimentation. Here are the outputted rules corresponding to the guideline for imaging following head trauma. The errors (differences manually compared with the original guidelines) are marked in red.

Rule 1: IF Person 16 or over with head injury THEN Proceed to Rule 2
 Rule 2: IF Any high-risk factors for CT head scan? (GCS ≤ 12 initial, GCS < 15 at 2 hours, open/depressed skull fracture, basal skull fracture signs, post-traumatic seizure, focal neurological deficit, > 1 vomiting episode) THEN Outcome: Do CT head scan within 1 hour ELSE Proceed to Rule 3
 Rule 3: IF Loss of consciousness or amnesia since injury? THEN Proceed to Rule 4 ELSE Proceed to Rule 5
 Rule 4: IF Any moderate risk factors? (age ≥ 65 , bleeding/clotting disorder, dangerous mechanism, > 30 minutes retrograde amnesia) THEN Outcome: Do CT head scan within 8 hours or within 1 hour if presenting > 8 hours after injury ELSE Proceed to Rule 5
 Rule 5: IF Taking anticoagulants or antiplatelets (excluding aspirin monotherapy)? THEN Outcome: Consider CT head scan within 8 hours or within 1 hour if presenting > 8 hours after injury ELSE Outcome: CT head scan not needed

Figure 2: MiniGPT4 rules

Rule 1: IF Person 16 or over presenting to the emergency department with a head injury THEN Proceed to Rule 2
 Rule 2: IF a GCS score of 12 or less on initial assessment in the emergency department THEN Outcome: (no path) ELSE Proceed to Rule 3
 Rule 3: IF a GCS score of less than 15 at 2 hours after the injury on assessment in the emergency department THEN Outcome: (no path) ELSE Proceed to Rule 4
 Rule 4: IF suspected open or depressed skull fracture THEN Outcome: (no path) ELSE Proceed to Rule 5
 Rule 5: IF any sign of basal skull fracture (haemotympanum, 'panda' eyes, cerebrospinal fluid leakage from the ear or nose, Battle's sign) THEN Outcome: (no path) ELSE Proceed to Rule 6
 Rule 6: IF post-traumatic seizure THEN Outcome: (no path) ELSE Proceed to Rule 7
 Rule 7: IF focal neurological deficit THEN Outcome: (no path) ELSE Proceed to Rule 8
 Rule 8: IF more than 1 episode of vomiting THEN Outcome: (no path) ELSE Proceed to Rule 9
 Rule 9: IF Are they taking anticoagulants (including vitamin K antagonists, direct-acting oral anticoagulants, heparin and low molecular weight heparins) or antiplatelets (excluding aspirin monotherapy)? THEN Outcome: Consider a CT head scan within 8 hours ELSE Proceed to Rule 10
 Rule 10: IF Is there loss of consciousness or amnesia since the head injury? THEN Proceed to Rule 11 ELSE Outcome: (stop)
 Rule 11: IF age 65 years or over THEN Outcome: Do a CT head scan within 8 hours ELSE Proceed to Rule 12
 Rule 12: IF any bleeding or clotting disorders (liver failure, haemophilia, taking anticoagulants or antiplatelets) THEN Outcome: Do a CT head scan within 8 hours ELSE Proceed to Rule 13
 Rule 13: IF dangerous mechanism of injury (a pedestrian or cyclist struck by a motor vehicle, an occupant ejected from a motor vehicle, or fall from a height of more than 1 m or 5 stairs) THEN Outcome: Do a CT head scan within 8 hours ELSE Proceed to Rule 14
 Rule 14: IF more than 30 minutes' retrograde amnesia of events immediately before the head injury THEN Outcome: Do a CT head scan within 8 hours ELSE Outcome: (stop)

Figure 3: Llama rules

We then turn to vision models, with the idea that the decision logic of NICE guidelines is not only contained in their text, but also in their visual organisation. The main difference from the previous pipeline is that, instead of extracting raw text from the PDF, we first convert each guideline page into an image and pass it to a multimodal model. Again, the model is prompted to return a structured JSON object extracting guideline metadata, rules, nodes, and edges. In this step, we compare two models: MiniVision and LLaVA. They were chosen because they offered two practical multimodal baselines with different strengths: MiniVision for more stable structured extraction, and LLaVA as a widely used open vision-language model for image-based reasoning. Here are the outputted rules corresponding to the same guideline imaging following head trauma.

Rule 1: IF Person 16 or over presenting to the emergency department with a head injury THEN Proceed to Rule 2 ELSE Proceed to Rule 2
 Rule 2: IF Are any initial risk factors present? (GCS ≤ 12 initially; GCS < 15 at 2 hours; suspected open/depressed skull fracture; basal skull fracture signs; post-traumatic seizure; focal neurological deficit; > 1 episode vomiting) THEN Outcome: Do a CT head scan within 1 hour of any risk factor being identified and make a provisional written radiology report available within 1 hour of the scan ELSE Proceed to Rule 3
 Rule 3: IF Is there loss of consciousness or amnesia since the head injury? THEN Proceed to Rule 4 ELSE Proceed to Rule 5
 Rule 4: IF Are any secondary risk factors present? (Age ≥ 65 ; bleeding/clotting disorders; dangerous mechanism; > 30 mins retrograde amnesia before injury) THEN Outcome: Do a CT head scan within 8 hours of injury or within 1 hour if presenting > 8 hours after injury and make a provisional written radiology report available within 1 hour of the scan ELSE Outcome: CT head scan not needed
 Rule 5: IF Are they taking anticoagulants or antiplatelets (excluding aspirin monotherapy)? THEN Outcome: Consider a CT head scan within 8 hours of injury or within 1 hour if presenting > 8 hours after injury and make a provisional written radiology report available within 1 hour of the scan ELSE Outcome: CT head scan not needed

Figure 4: MiniVision rules

Rule 1: IF Person 16 or over presenting to the emergency department with a head injury THEN Proceed to Rule 2
 Rule 2: IF Any of these risk factors are present (a GCS score of 12 or less on initial assessment, a GCS score of less than 15 at 2 hours after the injury, **suspected skull fracture**, sign of basal skull fracture, post-traumatic seizure, focal neurological deficit, more than 1 episode of vomiting) THEN **Outcome**: Do a CT head scan within 1 hour ELSE Proceed to Rule 3
 Rule 3: IF Loss of consciousness or amnesia since the head injury THEN Proceed to Rule 4 ELSE Proceed to Rule 5
 Rule 4: IF Any of these risk factors are present (age 65 years or over, bleeding or clotting disorder, dangerous mechanism of injury, **more than 30 minutes' amnesia**) THEN **Outcome**: Do a CT head scan within 8 hours of the injury, or within 1 hour if presenting more than 8 hours after injury ELSE **Outcome**: CT head scan not needed
 Rule 5: IF Taking anticoagulants or antiplatelets THEN **Outcome**: Consider a CT head scan within 8 hours of injury ELSE **Outcome**: CT head scan not needed

Figure 5: LLaVA rules

At this stage, we have therefore achieved the first part of the transformation: moving from heterogeneous guideline documents to a structured set of explicit clinical rules that can be followed by practitioners. Among the tested models, and across our 10 guidelines, MiniVision provides the most faithful transformation of NICE guidelines into structured decision logic. Its extracted rules are the most complete and the closest to the source flowcharts, whereas the other models more often omit links, flatten branches, or require manual repair.

3.2 Rules to decision trees

The rule representation obtained in the previous step is already useful in itself, since it provides a structure that practitioners can in principle follow to reach a clinical decision. In that sense, the transformation could stop at the rule stage. We nevertheless choose to reconstruct the rules into decision trees, in a logic of proof by bijection. Indeed, if the extracted rules can be mapped back into a structure close to the original flowchart, this provides evidence that the transformation has preserved the underlying guideline logic. A tree representation also makes it easier to inspect correctness and completeness, since differences become visually apparent. Finally, they provide a cleaner format for tracing outputs back to the original guidelines, which is particularly important in a clinical setting where recommendations should remain transparent and auditable.

To perform this reconstruction, we use the representation produced during rule extraction: the set of condition and action nodes together with the directed edges linking them. We then enumerate all possible traversals through this graph, starting from entry nodes and following outgoing edges until a terminal action is reached. This makes it possible to recover the full set of decision paths encoded by the guideline. The graph is then rendered with conditions as diamond-shaped nodes, actions as rounded boxes, and correct edge labels. In addition, individual traversals can be highlighted to visualise specific clinical pathways and verify that the extracted structure remains coherent from start to finish.

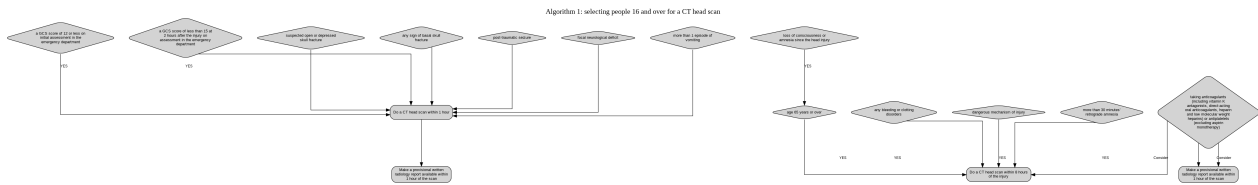


Figure 6: Llama based tree

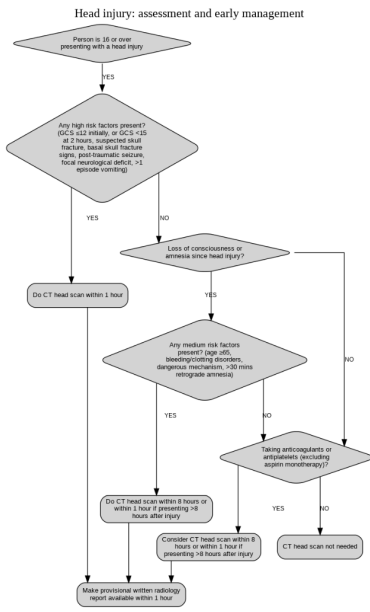


Figure 7: MiniGPT4 based tree

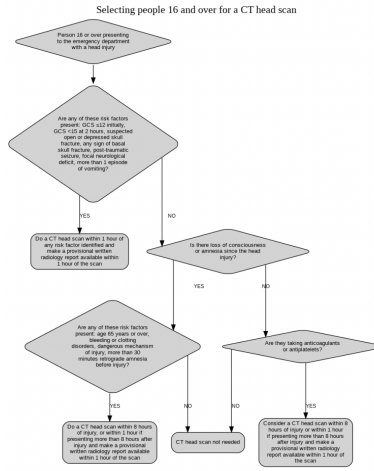


Figure 8: MiniVision based tree

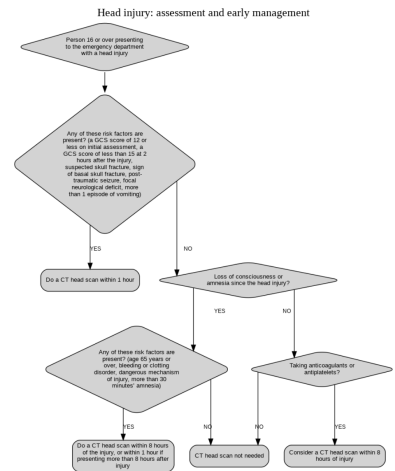


Figure 9: LLaVA based tree

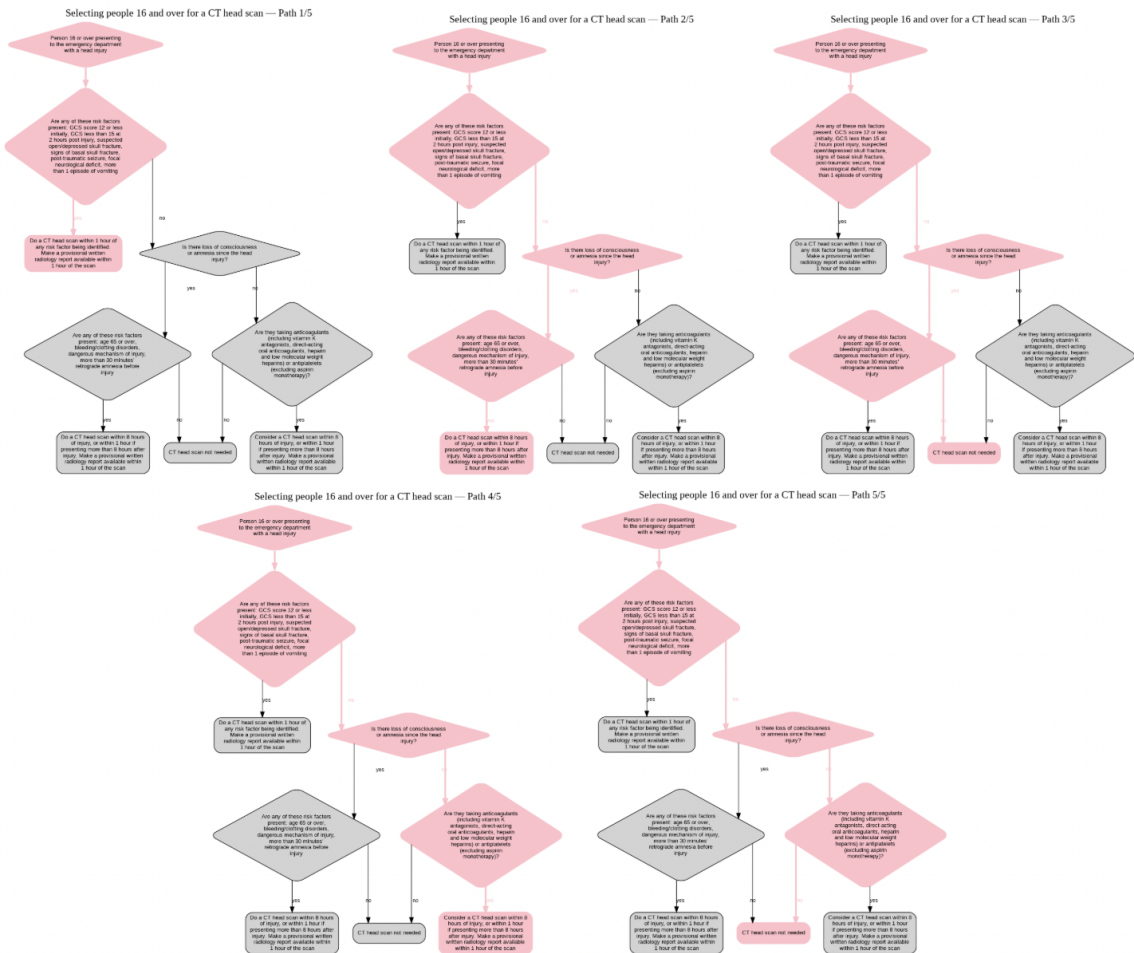


Figure 10: MiniVision based tree with highlighted paths

3.3 Assessing

At this stage, having obtained rule-based and decision tree-based representations for all models, we aim to determine which model performs best in a more quantifiable way. We also compare performance across different NICE guidelines, in order to assess whether the pipeline remains robust when applied to documents with different visual formats and levels of complexity.

For each of the ten NICE guidelines considered in this project, we evaluate the MiniVision extracted decision tree against the original guideline. This assessment is based on two criteria: the portion of correctly recovered nodes and edge labels, and the portion of correct placement of these nodes and labels. The first criterion checks if the model correctly extracts clinical information. And the second verifies that the decision logic is kept.

Guideline	Recovered nodes	Placed nodes	Placed edges	Comments
Hypertension	29/30	28/30	29/31	Comment boxes interfering with flowchart extraction
Pregnant women at risk of pre-eclampsia	4/5	5/5	3/4	1 missing condition
Preventing relapse in depression	17/17	17/17	22/22	
Antimicrobial prescribing in acute otitis media	15/15	13/15	16/17	Struggle with parallel actions
Antimicrobial prescribing in acute sore throat	10/10	10/10	12/12	
Management of chronic open angle glaucoma	28/30	28/30	28/33	Struggle with complex arrow scheme/placement; not handling two-way arrows
Management of ocular hypertension	23/23	23/23	21/23	Struggle with complex arrow scheme/placement; not handling two-way arrows
Antimicrobial prescribing in recurrent UTI	21/23	21/23	19/21	Struggle with parallel actions and simultaneous horizontal and vertical branches
Imaging following head trauma	33/33	33/33	41/42	
Antimicrobial prescribing human and animal bites	21/21	15/21	18/24	Struggle to represent table structure

Table 1: Assessment of extracted decision tree per guideline

We observe that performance is not consistent across all guideline types. The pipeline achieves near-complete recovery on more clearly structured flowcharts, such as “Imaging following head trauma”, “Antimicrobial prescribing in acute sore throat”, and “Preventing relapse in depression”, but performs less well on guidelines with more complex visual layouts. Lower placement and edge accuracy is particularly associated with comment boxes, dense grouping of conditions and actions, parallel actions, table-like structures, and complex arrow schemes, including two-way arrows and simultaneous horizontal and vertical branching. These cases highlight that guideline format itself remains an important source of difficulty for automated extraction.

4 Preparing extracted guideline trees for downstream use

This section extends the extracted decision trees into representations that are more directly useful for chatbot or clinical decision support applications. Rather than evaluating a deployed conversational system, we focus on preparing the guideline structures for downstream use.

4.1 Synthetic data creation

Testing databases are important in the development of health chatbots and overall clinical decision support systems. They help evaluate if a system behaves consistently across a wide range of patient situations. This is particularly important in medical settings. A structured testing database therefore provides a practical way to probe coverage, identify failure cases, and support later stages of validation.

To build such a resource, we use the extracted decision trees and traverse every complete path from an entry to an outcome. Each path is interpreted as defining one synthetic patient case compatible with the corresponding sequence of decisions. Meaning each path specifies the set of constraints that the patient attributes satisfy, and these constraints are instantiated into values consistent with the branch. Repeating this process for every path produces, for each NICE guideline in the project, a database of synthetic patients covering all reachable decision outcomes.

Each patient record contains a unique identifier, the final clinical outcome, the rule-path expressed in the rule-based format, and the medical attributes relevant to that guideline. Some attributes may remain empty, since not every branch requires all information.

Patient ID	Final outcome	Rule path	Age	Post-traumatic seizure	Amnesia since head injury	Taking anti-coagulants
1	Do a CT head scan within 1 hour of any risk factor being identified. Make a provisional written radiology report available within 1 hour of the scan	1 → 2		TRUE		
2	Do a CT head scan within 8 hours of injury, or within 1 hour if presenting more than 8 hours after injury. Make a provisional written radiology report available within 1 hour of the scan	1 → 2 → 3 → 4	72	FALSE	TRUE	
3	CT head scan not needed	1 → 2 → 3 → 4	40	FALSE	TRUE	
4	Consider a CT head scan within 8 hours of injury, or within 1 hour if presenting more than 8 hours after injury. Make a provisional written radiology report available within 1 hour of the scan	1 → 2 → 3 → 5		FALSE	FALSE	TRUE
5	CT head scan not needed	1 → 2 → 3 → 5		FALSE	FALSE	FALSE

Table 2: Extract of an example database for the NICE guideline on imaging following head trauma

In the current implementation, we generate one synthetic patient example for each complete decision path, building a dataset covering the full guideline. However, the approach can be extended. We can produce a larger and more diverse dataset by sampling multiple patient examples from the same path with different attribute values. And we can also remove selected attributes from otherwise valid cases to simulate situations with missing patient information. These extensions can be useful for future chatbot development, where robustness to variation and incomplete data is essential.

4.2 Handling missing patient information

In real clinical settings, patient information is often incomplete. Indeed, a clinician may not have access to all details required by a guideline, especially in fast-paced environments such as emergency care. It is therefore important that any future chatbot or decision support tool can operate in cases where some information is missing, rather than assuming a fully specified patient profile from the start.

In this project, we therefore enrich our decision tree structure: when a path is not fully traversed, we generate a partial decision tree starting from the LLM-reformulated clinical question to be asked and the paths the answers can lead to.

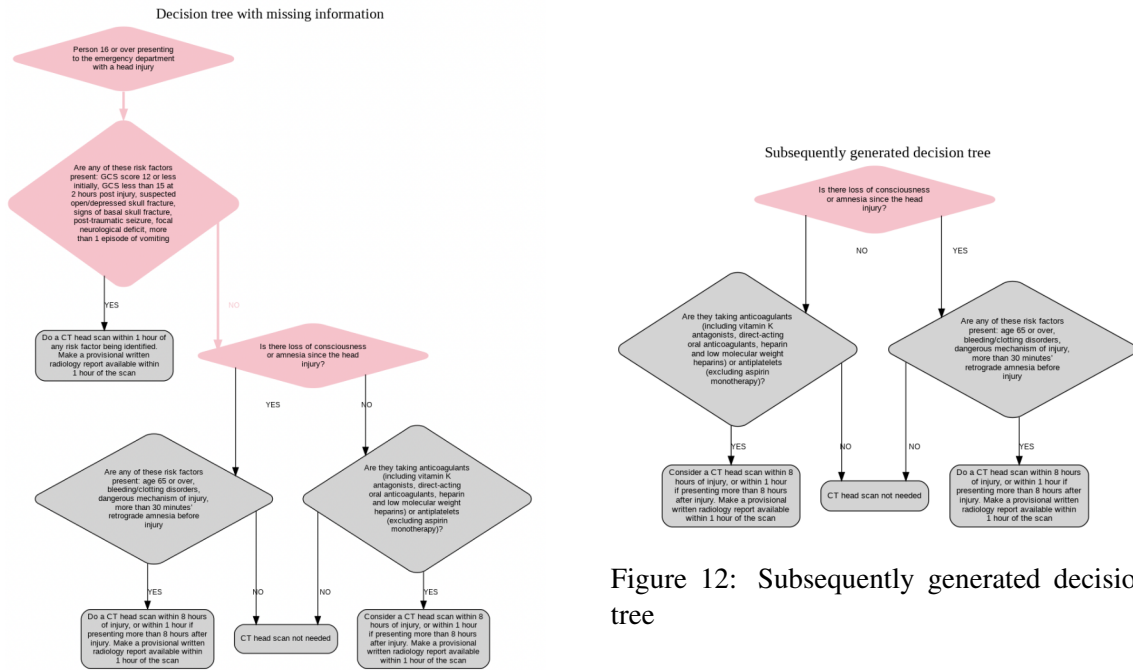


Figure 11: Decision tree with missing information

Figure 12: Subsequently generated decision tree

In this way, clinicians are given both the questions that can guide further assessment and the set of potential downstream decisions for the patient. This makes our structure better for conversational use, while keeping transparency about how each answer changes the clinical pathway.

4.3 Urgency assessment

Since the broader goal of guideline-based health chatbots and clinical decision support tools is to assist practitioners and make assessment easier and faster, it can be useful to add an indication of urgency to the extracted guideline structures. Indeed, not all possible outcomes of a clinical decision have the same emergency level. Adding an urgency marker therefore makes the output easier to interpret at a glance and can help highlight which situations deserve immediate attention.

To do this, we use LLMs to compare urgency and gravity of the different clinical outcomes. We then add comments to the final rules in the rule-based representation, and reflect this information visually in the decision trees by marking the leaf nodes with different colors.

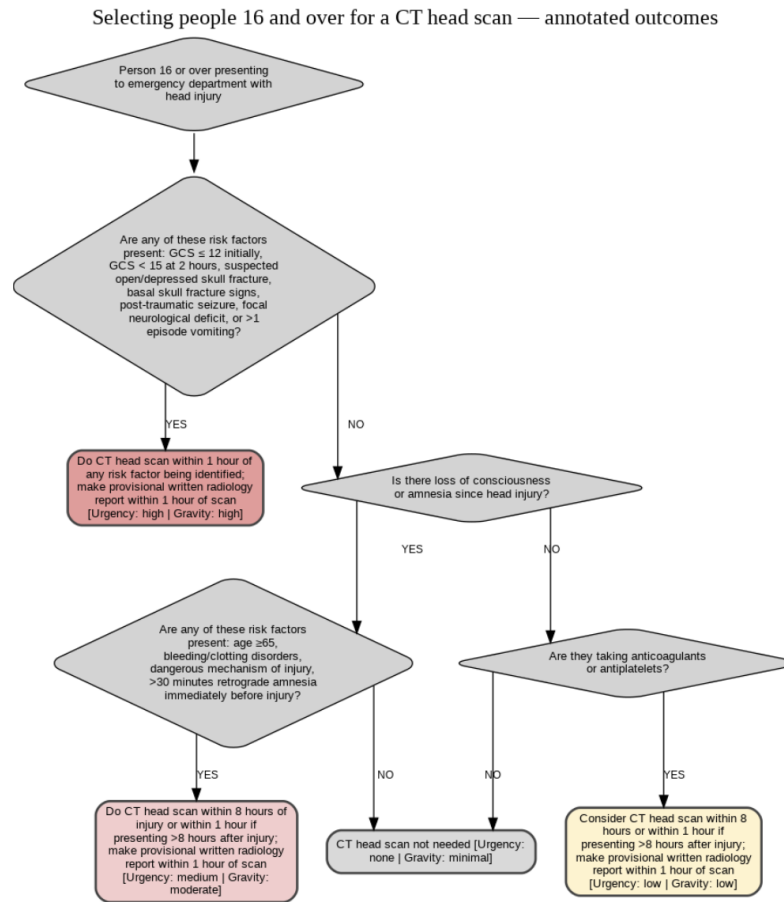


Figure 13: Marked decision tree

However, it is important to note that this is more interpretative than the rest of the pipeline. Unlike rule and decision tree extraction, urgency assessment does not simply recover information already explicit in the NICE guideline, but instead adds a generated judgement comparing outcomes independently of the original document structure. It should therefore be understood as an additional layer of derived information rather than as a direct extraction from the guideline itself.

5 Discussion

5.1 Main findings and relation with literature

The main finding of this project is that NICE clinical guidelines can be transformed into structured rule-based and decision tree-based representations while preserving a substantial part of their original clinical logic. Among the four models tested, MiniVision built the most faithful transformation, producing the most coherent rule sets and the most complete reconstructed trees.

These findings are consistent with previous literature showing that representation strongly affects the performance of language models on clinical guidance tasks. Kresevic et al. show that model accuracy improves when guideline content is cleaned, normalised, and converted into a more structured form, and our work supports this conclusion. Our results are also in line with MedDM, which treats flowchart structure as a central component of executable clinical reasoning and shows that tree-like representations are better than free-form responses to step-based medical decision support.

A further finding is that reconstructing decision trees from the extracted representation is useful not only as a visual output, but also for verification. By attempting to map extracted rules back into a structure close to the source flowchart, we are able to assess whether important branches, nodes, and transitions have been preserved. This bijective perspective strengthens the interpretability of the pipeline and helps identify where extraction errors occur. In addition, the resulting trees can be extended into resources for downstream work, including synthetic patient databases, support for incomplete-information cases, and urgency-marked outcomes.

5.2 Limitations

This project has several limitations. The main one is that the extraction models are imperfect. Our transformation pipeline is not flawless, and some of the observed errors are likely linked to the limits of the models used. It is plausible that stronger multimodal models, or models adapted more specifically to the structure of NICE guideline documents, would produce more accurate and more stable extractions. Moreover, we have observed that performance is not uniform across all guideline formats. This suggests that the models we used are still sensitive to document structure, and that guideline formatting still is an important obstacle for automated extraction.

Another limitation is the scale of the evaluation. Although the ten NICE guidelines considered here cover a variety of topics and layouts, they do not represent the full diversity of clinical guidelines encountered in practice. The results should thus be interpreted as encouraging rather than fully generalizable. In addition, even if the evaluation criteria used in this project provide a structured way to compare outputs, judging whether content and placements are correct can still involve some degree of manual interpretation.

Finally, the extracted decision trees remain simplified representations of clinical reasoning. Even when a NICE flowchart is captured accurately, real clinical decision-making may also depend on local protocols, patient history, and contextual judgement that are not fully represented in national guideline algorithms alone.

5.3 Future Research

A first direction for future research would be to test these structures in the chatbot itself. Access to the deployed system would make it possible to evaluate how well the extracted rules and decision trees support real conversational guidance, and to identify practical improvements in retrieval, questioning strategy, and response formatting. This would be an important next step, since this project prepares the structures for downstream use

but does not yet assess their performance in interaction.

A second direction would be to extend this work beyond the ten guidelines considered here and apply the same pipeline to a much larger set of NICE documents. This could make it possible to build a broader database of machine-readable guideline structures, including rules, decision trees, synthetic patient cases, and urgency-marked outcomes. Such a resource could then support future guideline-based chatbots or clinical decision support systems at a larger scale, while also enabling more systematic evaluation across medical domains and document formats.

A further line of research would be to improve the extraction pipeline itself. As suggested by the limitations above, different models might produce more reliable transformations, especially for visually dense or irregular documents. It would also be interesting to explore whether additional post-processing or hybrid extraction methods could improve robustness while preserving the transparency of the rule- and tree-based representations.

6 Conclusion

This study aimed to investigate how NICE clinical guidelines can be transformed into formal structures suitable for computational use while preserving their original clinical reasoning. To do so, we extracted rule-based representations from guideline documents, reconstructed them into decision trees, and assessed the consistency of this transformation across several models and guideline formats. Among the tested approaches, MiniVision provided the most faithful transformation, producing the most complete and coherent structured representations. Beyond the extraction itself, the resulting trees were also extended into resources for downstream use, including synthetic patient databases, support for missing patient information, and urgency-marked outcomes. Overall, this work shows that NICE guidelines can be converted into interpretable machine-readable decision structures, and that such representations can provide a useful basis for future guideline-based clinical decision support and conversational systems.

7 Acknowledgments

I would like to express my sincere gratitude to my advisor, Professor Julia Ive, for her guidance, valuable feedback, and continuous support throughout this research project. I had the best time working and exchanging with her.

I am also grateful to University College London for providing an excellent academic environment and the resources necessary to conduct this work.

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A Appendix

A.1 Prompt templates used in the guideline extraction pipelines

This appendix contains the prompt templates used in the proposed pipeline. The first prompt defines the schema for extracting clinical guideline flowcharts into a structured JSON decision graph, while the second prompt converts condition nodes into machine-readable evaluators for downstream rule processing.

```

SYSTEM_PROMPT = r"""
You are extracting a medical guideline into a STRICT JSON object that MUST conform to this schema:

GuidelineSchema = {
  guideline_id: string,
  name: string,
  version: string,
  citation: string,
  citation_url: string,
  rules: string[],
  nodes: Array<{
    id: string,
    type: "condition" | "action",
    text: string
  }>,
  edges: Array<{
    from: string,
    to: string,
    label: string
  }>
}

All fields are REQUIRED.
- If a value is unknown, use an empty string "" (or empty array [] where appropriate).
- Return EXACTLY ONE JSON object matching the schema above.
- Do not include any extra top-level keys.
- Do not include markdown, commentary, or explanation – JSON ONLY.

Extraction rules:
- rules: an array of explicit IF-THEN rule strings derived from the guideline (one rule per item).
  Example format: "IF <condition> THEN <action>".
- nodes: represent the guideline as a decision graph.
  - type="condition": a branching question/criterion (e.g., thresholds, presence/absence, ranges).
  - type="action": a recommendation, management step, or outcome.
  - text: concise natural-language description taken from the guideline.
  - id: unique stable identifier (e.g., "c1", "c2", "a1"...). IDs MUST be unique.
- edges: directed links between node ids.
  - from/to MUST reference existing node ids.
  - label is REQUIRED for every edge; use "" if no label is needed.
  - If the guideline indicates YES/NO branches, use labels like "yes" / "no".
  - If the guideline indicates categories (e.g., "stage 1", "stage 2"), put that in label.

Use the PDF via file_search. Prefer the guideline's own wording.
If parts are ambiguous, choose a conservative, minimal graph and keep wording generic (but still valid JSON).

Return JSON ONLY.
"""

```

```

EVALUATOR_PROMPT = r"""
You are converting condition nodes into machine-readable evaluators.

Input: you will receive JSON:
{
  "nodes": Array<{ "id": string, "type": "condition"|"action", "text": string }>
}

Output: Return STRICT JSON ONLY:
{
  "condition_evaluators": {
    "<condition_node_id>": ConditionEvaluator
  }
}

Only produce evaluators for nodes where type == "condition".
Do NOT produce evaluators for action nodes.

ConditionEvaluator is a recursive union of these forms:

1) Boolean flag:
{ "type": "bool", "variable": string }

2) Numeric comparison:
{
  "type": "numeric_compare",
  "variable": string,
  "op": ">=" | ">" | "<=" | "<" | "=",
  "threshold": number,
  "unit": string
}

3) Numeric range (inclusive):
{
  "type": "numeric_range",
  "variable": string,
  "min": number,
  "max": number,
  "unit": string
}

4) Count comparison:
{
  "type": "count_compare",
  "variable": string,
  "op": ">=" | ">" | "<=" | "<" | "=",
  "threshold": number
}

5) Enum membership:
{
  "type": "enum_in",
  "variable": string,
  "any_of": string[]
}

6) Logical OR:
{ "type": "or", "conditions": ConditionEvaluator[] }

7) Logical AND:
{ "type": "and", "conditions": ConditionEvaluator[] }

Mapping guidance (important):
- Prefer numeric_compare / count_compare when the text contains a clear threshold (e.g., "GCS <= 12", "more than 1 episode", "age 65 years or over").
- Use unit for numeric_compare/numeric_range. Examples:
  - GCS: unit=""
  - age: unit="years"
  - time: unit="minutes" or "hours"
- For phrases like "loss of consciousness or amnesia", use OR with two bool variables.
- For lists like "dangerous mechanism of injury (... list ...)", use enum_in with variable like "injury_mechanism" and any_of containing the listed categories.
- When the text is ambiguous, use a boolean flag rather than guessing numbers.

Variable naming:
- Use snake_case, concise, reusable (e.g., "gcs_initial", "gcs_2h", "vomiting_episodes", "age", "retrograde_amnesia_minutes").

Return JSON ONLY. No extra keys. No commentary.
"""

```

A.2 Extracted decision trees

This appendix presents the ten decision trees extracted from the NICE guideline visual summaries used in this study.

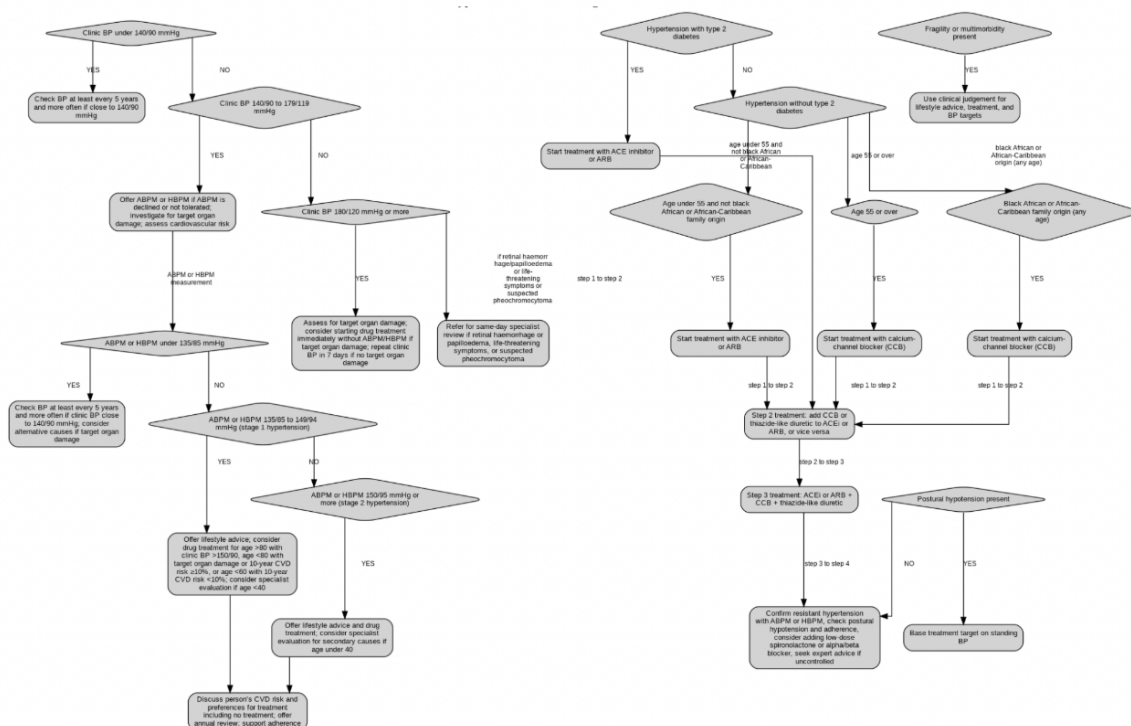


Figure 14: Hypertension in adults: diagnosis and treatment

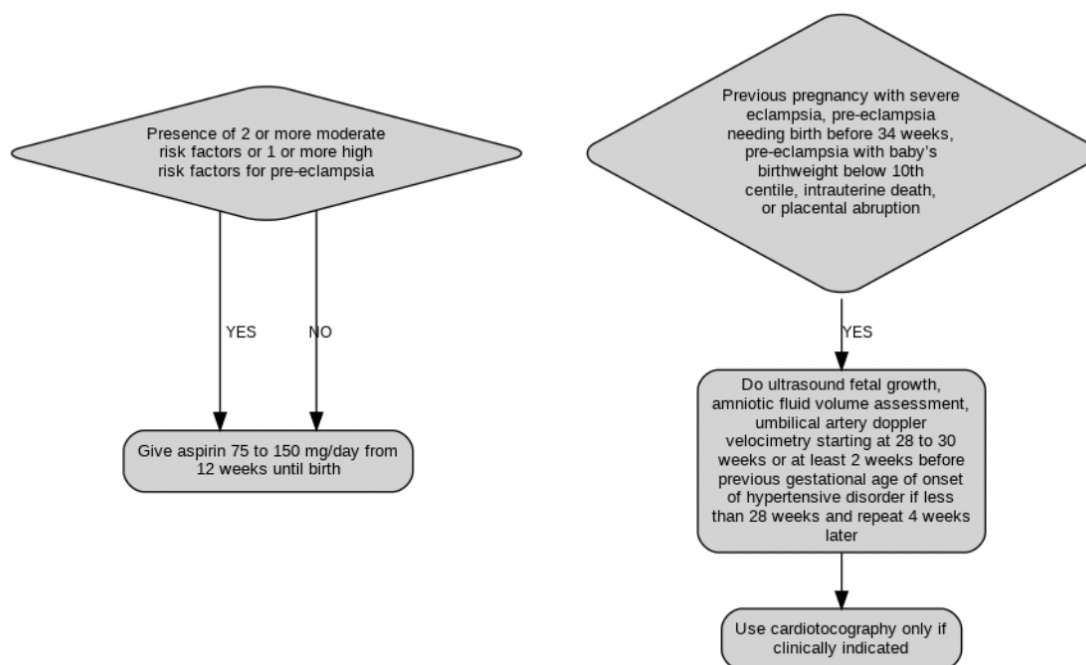


Figure 15: Pregnant women at risk of pre-eclampsia

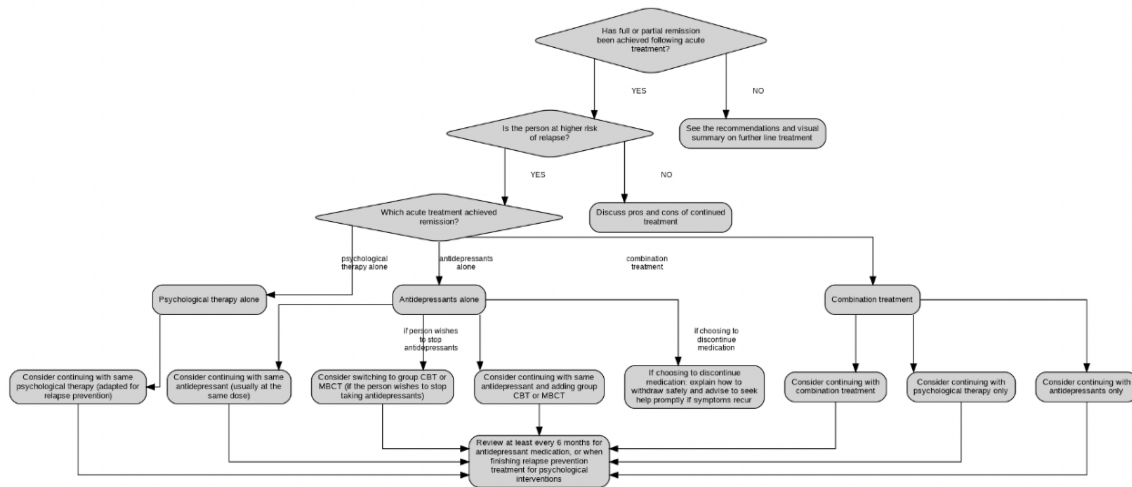


Figure 16: Preventing relapse in depression

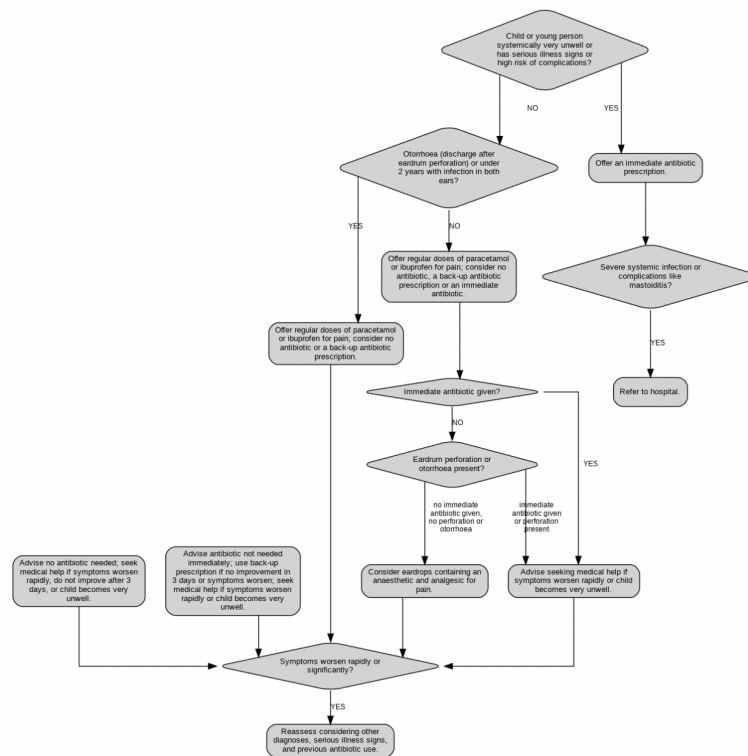


Figure 17: Antimicrobial prescribing in acute otitis media

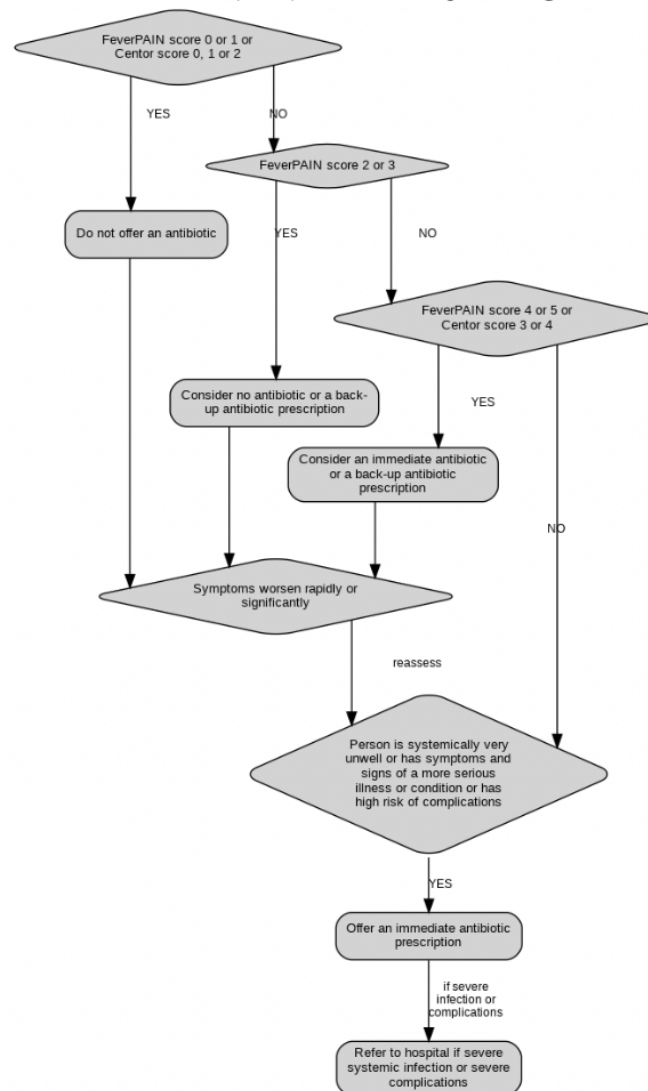


Figure 18: Antimicrobial prescribing in acute sore throat

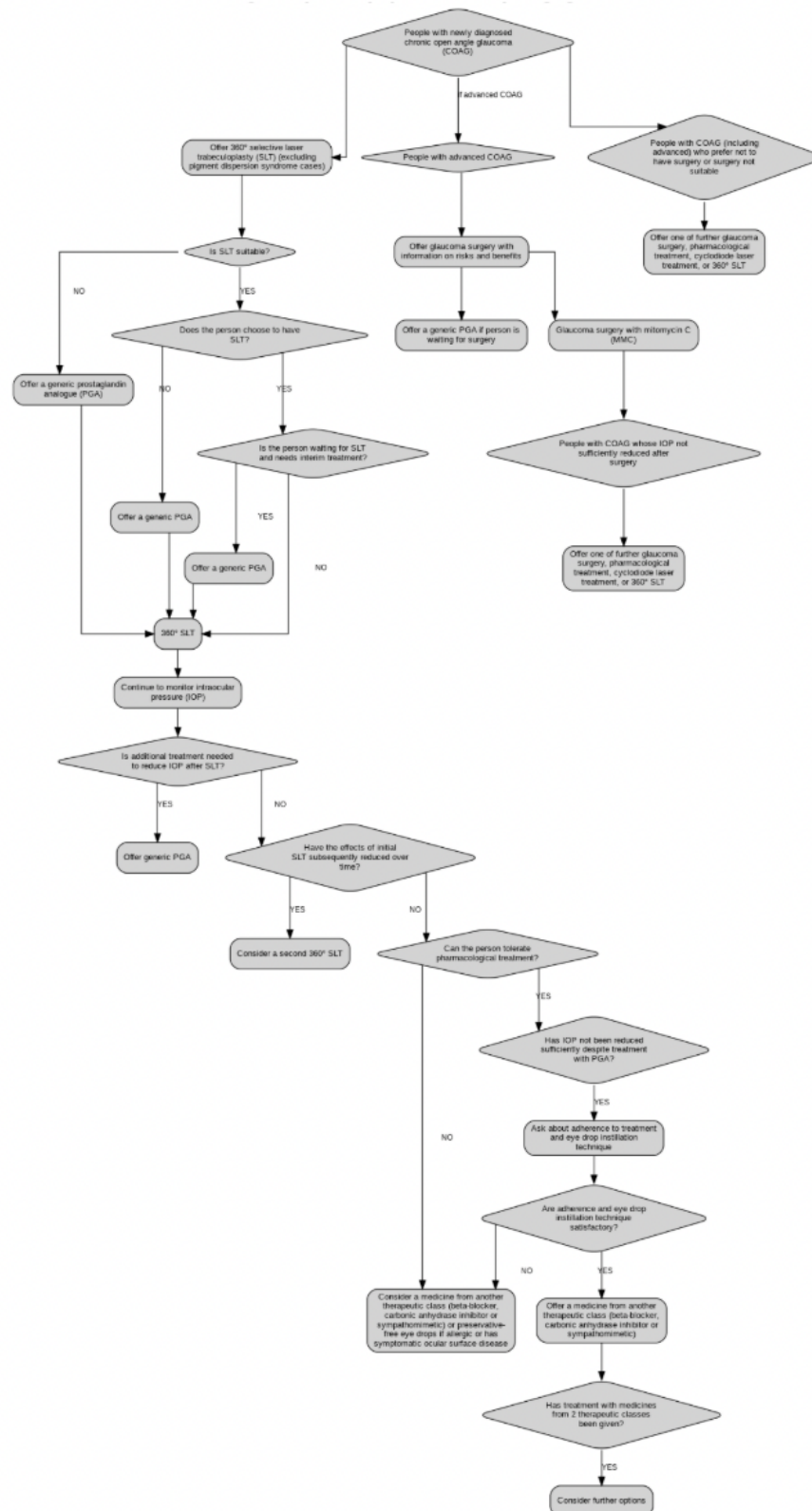


Figure 19: Management of chronic open angle glaucoma

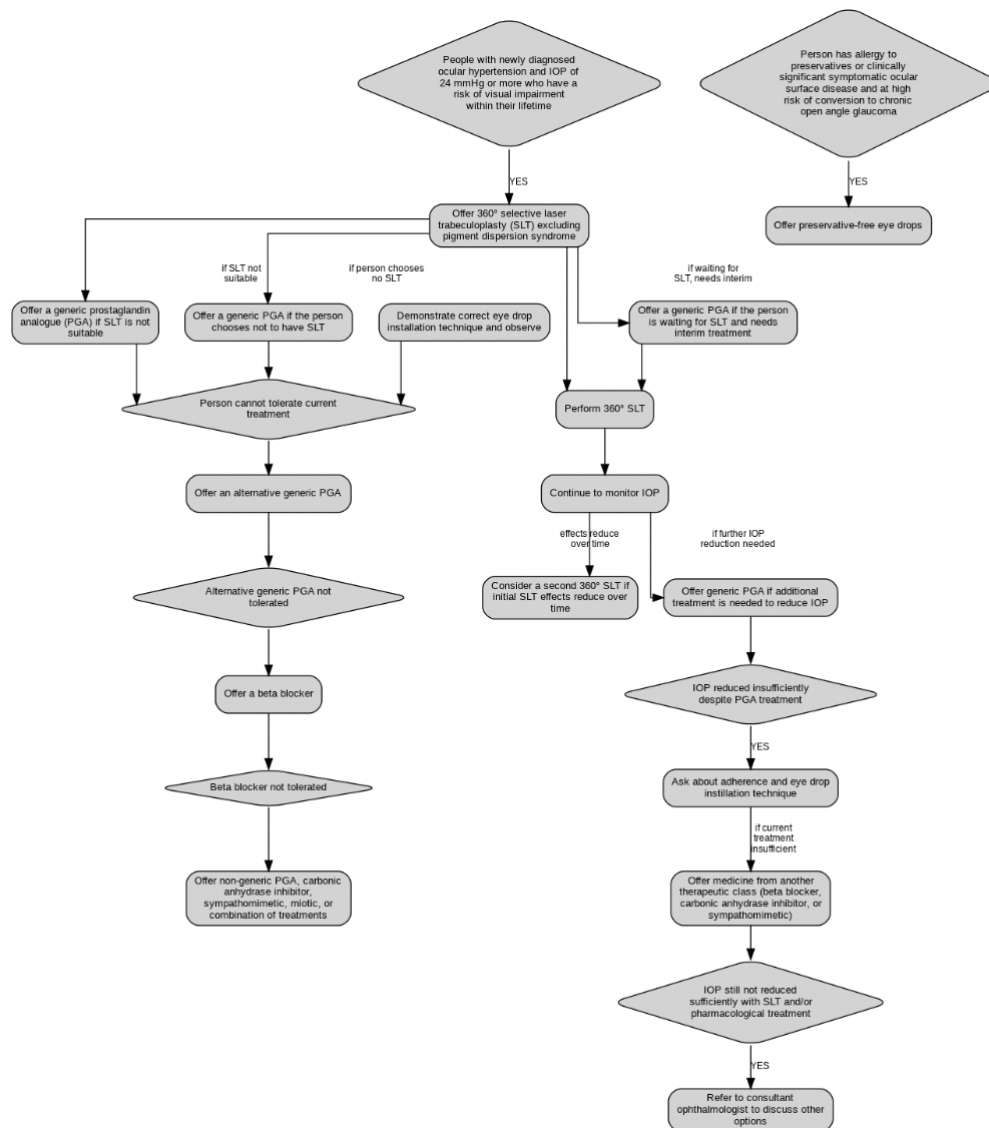


Figure 20: Management of ocular hypertension

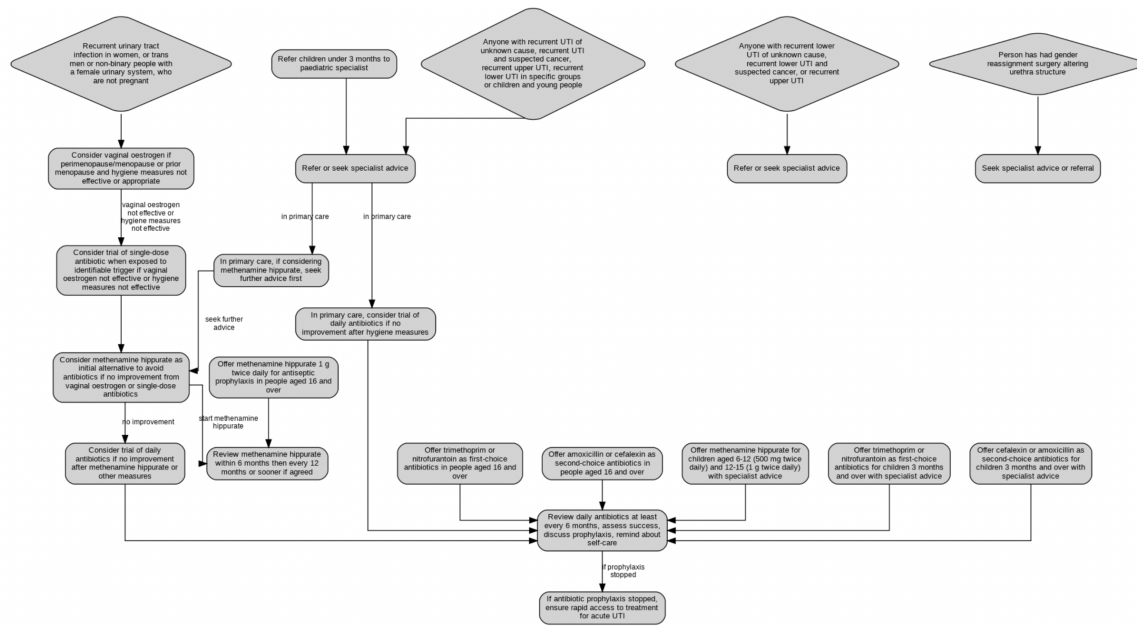
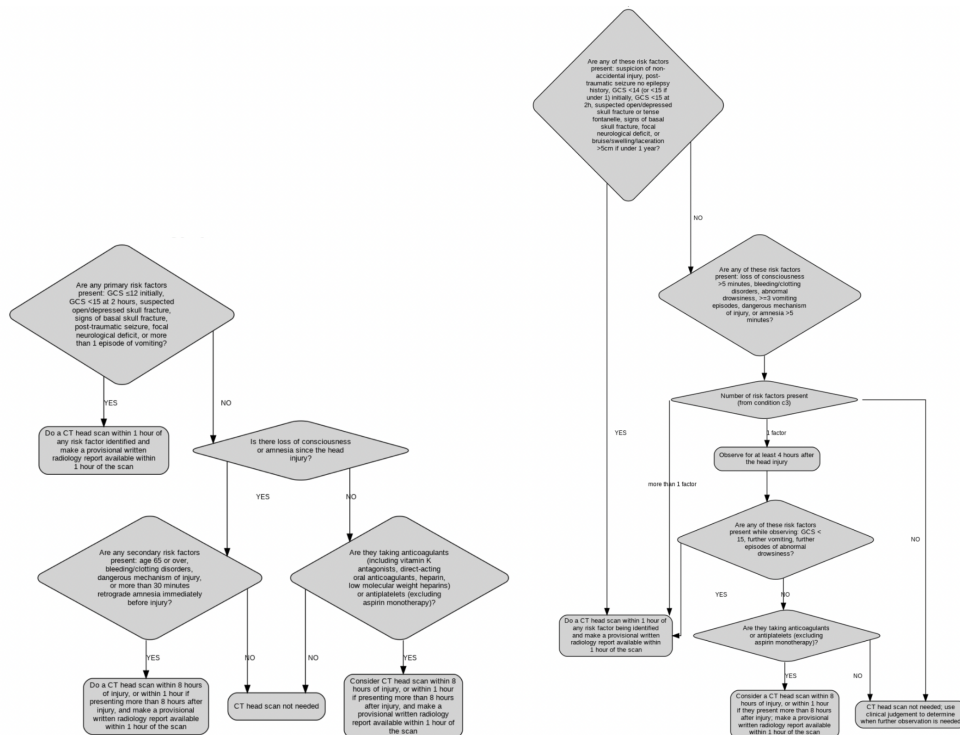


Figure 21: Antimicrobial prescribing in recurrent UTI



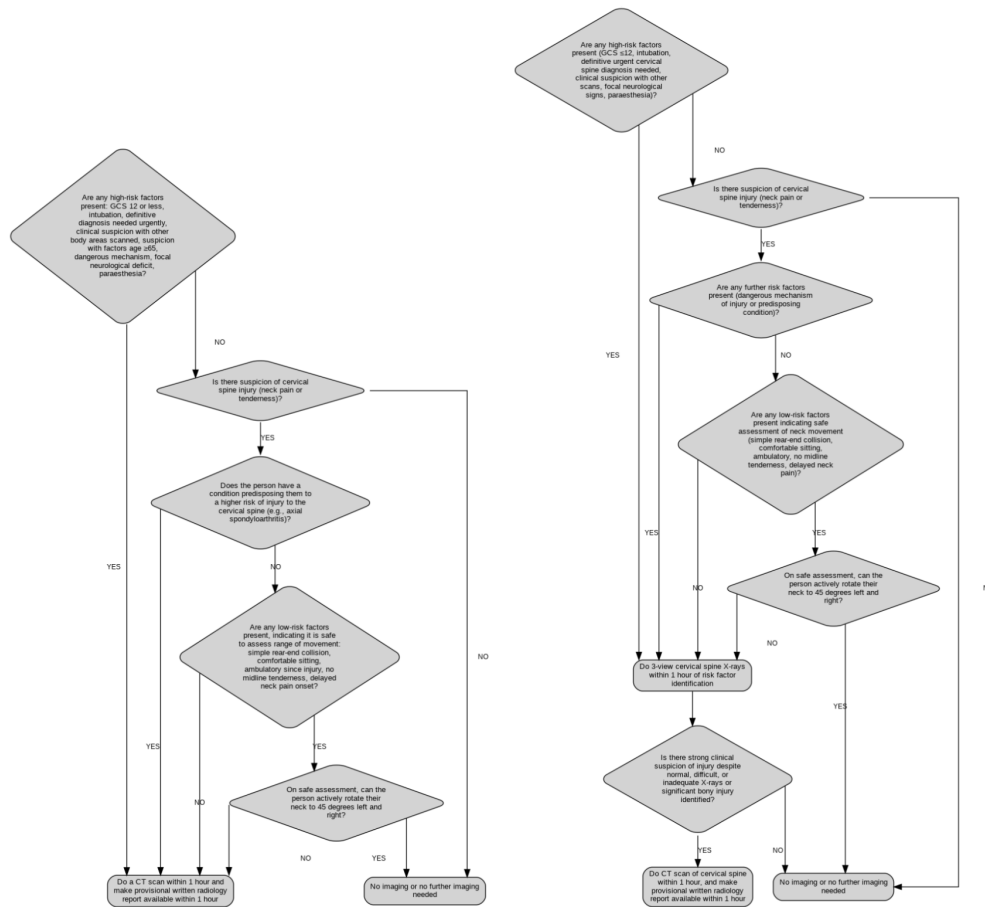


Figure 22: Imaging following head trauma

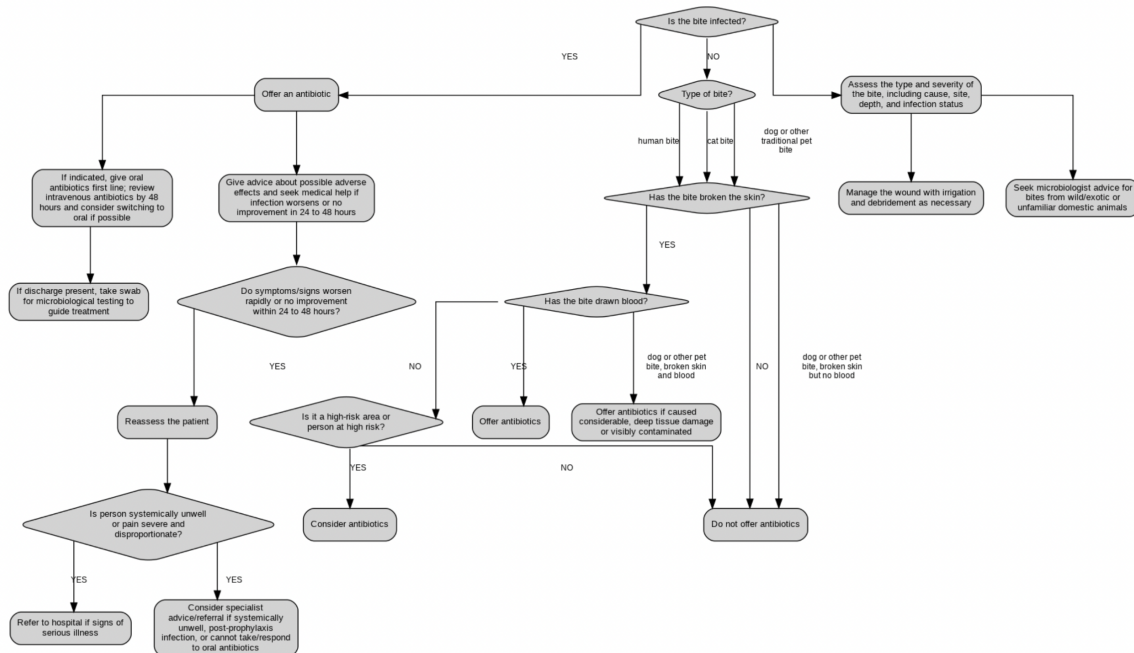


Figure 23: Antimicrobial prescribing in human and animal bites



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